National University of Computer and Emerging Sciences
Karachi Campus

Recommender System (CS4053)

Sessional-II Exam

Date: 4/7/2025

Course Instructor

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Total Time: 1hr
Total Marks: 30
Total Questions: 03

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Roll No Section Student Signature

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Attempt all questions.

CLO # 1: Describe different techniques in making personalized recommendations in various scenarios.

Q1: Provide brief answers to the following: **[Est. Time: *10 minutes*, Points: 6]**

a) Does *RippleNet* offer serendipity? If yes, then how can it be controlled?
Yes. It can be controlled by changing the hop-count or number of ripples.

b) Can content-based filtering work when we have high correlation between features? Justify your answer.
The correlation itself won't be affecting the algorithm. Although the highly correlated features can be removed to allow less resource consumption.

c) Can Matrix Factorization give zero recommendations? Justify your answer.
It cannot give zero recommendations as even in the case of high sparsity, only the accuracy and performance of the model suffers.

d) What is the benefit of using structural embedding in a graph-based recommender system?
It allows for extracting and use relationships and connections between items and users to improve the recommendations.

e) What is the major disadvantage of path-based knowledge graph-based systems?
For more complex and large graphs, it will not be feasible for use in terms of resource requirements.

f) Can a knowledge-based system ever be affected by lack of serendipity? Justify your answer.
A knowledge-based system always considers user requirements and offers highly personalized recommendations. It will not contain serendipity due to its primary objective of catering to current user requirements.

CLO # 2 Solve mathematical optimization problems pertaining to recommender systems.

Q2: Consider the given interaction matrix ***R*** and its factors ***U*** and ***V***. **[Est. Time: *25 minutes*, Points: 12]**

5	5	5	0	0	0
5	4	5	0	0	0
1	5	2	0	0	0
5	5	3	5	5	5
-1	-4	-5	1	1	1
-2	-1	-1	1	1	1
-3	-5	-1	1	1	1

5	1
-2	1
-4	0
1	0
0	1
0	0
0	0

1	-3	-2	-5	-1	-5
0	-2	0	1	0	-6

a) If r and c represent rows and columns, do the factors contain valid values for original interactions in cells ***R*(row1, col2)** and ***R*(row5, col1)**?
Invalid values for both R(row1, col2) and R(row5, col1)

b) The rating system allows positive ratings (likeness), negative ratings (dislike) and rating value *0* (neutral). How can we transform this into a rating system that is in range [1-5] such that users' original opinions towards items are preserved?
To transform this to a rating system in range [1-5], we can make a business rule to consider all ratings between 1-2 to be negative, ratings between 4-5 to be positive, while the rating 3 can be assumed to represent a neutral opinion.

c) For what value of ***k*** will any two given factors be consuming more storage units than the original interaction matrix 7x6 matrix ***R***?
For any value of k that is equal or greater than 4, two given factors will consume more memory than the original interaction matrix.

d) Explain how Matrix Factorization is affected by cold-start users.
For a cold-start user, a new row will be added to the m x k factor. This new row will only be passively trained from the other values in the factor and therefore will not be accurately predicting the ratings for this user.

CLO # 2 Solve mathematical optimization problems pertaining to recommender systems.

Q3: Consider the given table depicting features of laptops available for purchase:

	Resolution (in thousands)	Battery (in hours)	Precision (in units)	Shutter speed (in seconds)	Price
Camera 1	**9.21**	**3**	**56**	**30**	**1,69,000**
Camera 2	**8.54**	**2.5**	**81**	**25**	**1,72,000**
Camera 3	**6.82**	**4**	**80**	**45**	**1,71,500**

	Resolution (in thousands)	Battery (in hours)	Precision (in units)	Shutter speed (in seconds)	Price
Requirements	**8.75**	**6**	**52**	**35**	**1,51,000**
Weights	**1**	**1**	**1.2**	**1.5**	**1**

a) Use case-based knowledge driven technique to recommend the most suitable camera for the given user requirements.

Sim (U_{REQ}, Camera 1) = 0.90
Sim (U_{REQ}, Camera 2) = 0.74
Sim (U_{REQ}, Camera 3) = 0.75
Camera 1 is the closest to given user requirements.

b) Consider a scenario where we assign different weights to constraints in a constraint-based system to deal with zero recommendations such that when enough constraints are satisfied, a threshold value is reached, and recommendations are made. What can be the major drawback of this approach?

- Assigning and updating weights to constraints would require domain expertise
- Loss of personalization
- Low quality recommendations

[Est. Time: *20 minutes*, Points: 12]

Good luck